Scope and Evidence Register

Separate the document delivered today from implementation work proposed next

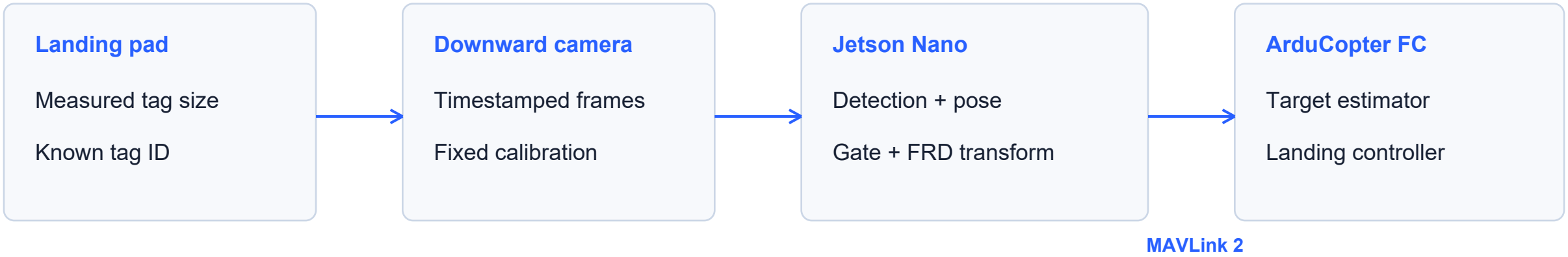
Workstream	Present evidence	Planned proof
Engineering design	This specification and illustrative calculations	Client review of assumptions and interfaces
Camera / AprilTag	Reference photo only; no calibration/test dataset	Calibration files; labeled positive and negative clips
Jetson software	No executable implementation delivered here	Source, dependency lock, replay and bench records
ArduPilot integration	No connection or parameter dump supplied	Pinned firmware; packet capture; target response
Flight performance	No UAV logs or landing trials supplied	Pilot-approved trials, video and measured touchdown error

Included proposal: fixed downward camera, one stationary AprilTag, healthy ArduCopter multirotor and supervised landing. Excludes moving decks, obstacle avoidance, SLAM, GPS-denied navigation, automatic launch and production certification.

Decision: Treat every numeric threshold in this document as proposed until a test report marks it passed.

System Architecture

Figure 1 - Proposed data and control paths; schematic, not an integrated system



Independent flight inputs

GPS/EKF, IMU and altitude/range sensing remain flight-controller inputs. Vision is a target-relative measurement.

Operator and evidence

RC takeover + ground station; companion CSV, MAVLink capture, autopilot .BIN and synchronized video.

Decision: Do not substitute tag detection for a valid horizontal navigation estimate. [1, 2]

Hardware and Operating Envelope

Candidate bench configuration - final part selection requires client hardware confirmation

Component	Proposed specification / verification
Compute	Jetson Nano 4 GB developer platform; regulated supply, cooling, storage and measured thermal headroom.
Camera	Downward USB UVC camera; 640 x 480 at 30 fps initial mode; fixed focus and exposure limits. Global shutter preferred.
Flight controller	ArduCopter-compatible controller with accessible telemetry port; exact board and firmware recorded before setup.
Target	tag36h11, proposed ID 0, 0.30 m detection-edge side length; flat matte print with white margin. Measure actual print.
Power / link	Separate regulated Nano power; common signal ground; verify UART voltage and pinout. Never power Nano from telemetry rail.
Initial envelope	Stationary level pad, daylight without glare; 1-3 m camera height, tilt <=10 degrees, wind <=2 m/s; proposed limits.

Decision: Confirm payload, camera visibility, electrical compatibility and pilot access before integration.

Camera Calibration and Target Geometry

Calibration is an engineering deliverable, not a default camera setting

Calibration procedure

Capture at least 25 sharp checkerboard views spanning image center, edges and varied tilt at the exact operating resolution.

Fit K and distortion coefficients; retain raw images, board dimensions, residuals and independent held-out images.

Proposed gate: held-out reprojection RMS <=0.8 px. Recalibrate after focus, lens, mount or capture-mode changes.

Detection geometry

Use AprilTag 3 with one allowed tag family and an ID whitelist. Reject edge-cropped tags and mismatched IDs.

Tag size means the distance between detection corners, not the whole paper sheet. Scale errors directly bias range. [4]

Resize/crop requires matching intrinsics: scale fx, fy, cx, cy; subtract crop origin from principal point. Avoid undocumented resizing.

Decision: Freeze camera mode and measured tag dimensions in a versioned calibration bundle.

Pose Estimation . From Pixels to Meters

Square-marker PnP with explicit corner ordering and ambiguity rejection [5]

Mathematical contract

Camera optical frame C: x right, y down, z along the optical axis. PnP returns tag-to-camera rotation R_CT and translation t_CT.

p_C = R_CT p_T + t_CT
For the tag center p_T = 0, its camera-relative position is t_CT. This is not the camera world pose.

IPPE_SQUARE object points, in meters:
(-s/2,+s/2,0), (+s/2,+s/2,0),
(+s/2,-s/2,0), (-s/2,-s/2,0).

Implementation checks

Map detector corners to that exact physical order; verify with a rotated tag. Use solvePnPGeneric to evaluate planar candidate poses.

Reject candidates behind camera; compare reprojection error and temporal continuity. If plausible candidates disagree materially, suppress output.

Use raw corners with K + distortion, or undistorted corners with matching K and zero distortion. Never compensate distortion twice.

Compare estimated range and lateral position with a measured bench grid; low residual alone does not prove metric accuracy.

Decision: Resolve scale, point ordering and planar ambiguity before enabling the telemetry publisher.

Coordinate Transform . Camera to Body FRD

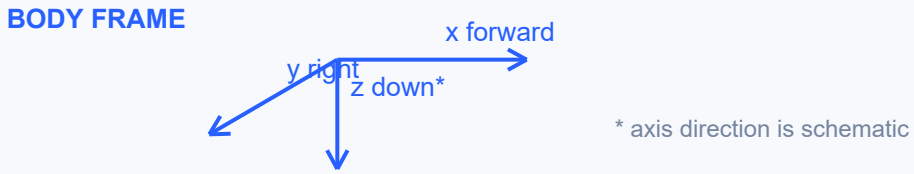
Figure 3 - Illustrative transform and sign test; actual extrinsics must be measured

Frame definition and transform

Body B: x forward, y right, z down. R_BC rotates camera coordinates into body coordinates. t_BC is camera origin relative to vehicle reference.

p_B = R_BC t_CT + t_BC
Example mount: camera x points body right; camera y points body aft; camera z points down.

R_BC = [[0,-1,0], [1,0,0], [0,0,1]]
Thus camera (0.10,-0.20,2.00) m maps to body (0.20,0.10,2.00) m if origins coincide.



Offset ownership and physical proof

Proposed publisher sends a vector from the vehicle reference, including measured lever arm once. Confirm firmware reference conventions on bench.

For this convention, keep PLND_CAM_POS_* zero and PLND_YAW_ALIGN zero to avoid duplicate correction. Verify on the pinned firmware.

Move target forward: x positive. Move right: y positive. Below aircraft: z positive. Rotate aircraft 90 degrees and repeat.

Camera yaw/lever-arm compensation must have one owner. Never transform to FRD and then apply the same mount correction again.

Decision: Fail integration if any axis sign or offset ownership check is ambiguous.

MAVLink LANDING_TARGET . Interface Contract

MAVLink 2 positional path; proposed valid-measurement rate 20 Hz [1, 3]

Field / transport	Proposed value	Interpretation
frame / position_valid	MAV_FRAME_BODY_FRD (12) / 1	Fresh validated body-relative position
x, y, z	p_B components in meters	Forward, right, down; finite and z > 0
distance	sqrt(x*x + y*y + z*z)	Populate positive target distance for ArduPilot
time_usec	Monotonic capture time, microseconds	Record send time separately; clock epoch documented
angle_x/y, size_x/y	0 in selected positional path	Do not silently switch to angle-only fallback
target_num, q, type	0; [1,0,0,0]; VISION_FIDUCIAL	Metadata; no yaw-control claim
Connection	MAVLink 2 serial or USB; unique component ID	Verify extension fields survive packet decode

Loss rule: stop publishing on stale/invalid detection. position_valid=0 selects an angle-based interpretation; it is not a safe generic "target lost" flag. Never send zero position as a placeholder or repeat old frames to imitate 20 Hz.

Decision: Verify received field values and actual estimator response; packet transmission alone is insufficient.

Software and Setup . Reproducible Installation

Implementation plan - commands and dependency pins are to be delivered after hardware validation

Runtime and modules

Candidate Nano baseline: NVIDIA JetPack 4.6.2 / L4T 32.7.2, which supports Nano. Confirm board variant and working image before dependency selection. [6]

Use system-compatible Python, OpenCV, AprilTag 3 and pymavlink; pin working versions/commits. CPU detection baseline; benchmark before optimization.

Planned modules: capture, calibrate, detect, pose, transform, quality gate, publish, logger, replay and health monitor.

Latest-frame queue of depth 1; monotonic timestamps; bounded processing. Log dropped frames and prevent stale backlog.

Commissioning sequence

1. Image Nano; verify power/cooling; record OS and library manifest. Confirm capture mode and calibration checksum.

2. Run offline replay and measured-grid pose checks. Verify transform with physical sign tests.

3. Props removed: connect telemetry, confirm heartbeat and MAVLink 2 decoding. Configure unique sysid/compid and link permissions.

4. Start service in observation mode. Enable publishing only after configuration validation; supervise startup, reconnect and shutdown tests.

Decision: Deliver a tested setup guide for the actual Nano image; do not promise compatibility from a generic install list.

Autopilot Setup . Proposed Configuration

Review against the selected firmware parameter metadata before applying [1, 2]

Parameter / check	Starting intent	Verification required
PLND_ENABLED / PLND_TYPE	1; reboot / 1 (MAVLink)	Refresh parameters and confirm source
Telemetry SERIALn_PROTOCOL	2 (MAVLink 2), port-specific	Confirm actual port mapping and baud
PLND_YAW_ALIGN / CAM_POS_*	0 / zero for page 8 convention	No duplicate rotation or lever arm
PLND_STRICT / retry policy	Candidate strict mode 2	Test retry climb, final hover and pilot recovery
PLND_ALT_MIN / ALT_MAX	Select with pilot after SITL	Document low-altitude loss and search behavior
PLND_XY_DIST_MAX / landing speed	Select to fit safe pad envelope	Prevent descent before lateral alignment
PLND_LAG / offsets / options	Measure latency; stationary target	Check available units and default behavior

Maintain a valid position/attitude estimate. First confirm disarmed target response, then stable Precision Loiter before landing. Target loss does not inherently abort or hold; retry, low-altitude descent and other failsafes require explicit verification.

Decision: Export before/after parameters and exercise target loss on the exact firmware before flight.

Acceptance Criteria . Proposed Pass Gates

All outcomes pending - thresholds require client agreement and measured evidence

Gate	Proposed threshold	Evidence / method
Calibration	Held-out RMS <=0.8 px	Independent board views; calibration report
Bench pose	95th-percentile lateral error <=0.05 m; range error <=5%	>=30 measured poses, 1-3 m range and tilt <=10 deg
Detection	>=95% valid frames with fully visible tag	>=60 s per lighting/height cell; log all rejects
False targets	0 accepted wrong/no-tag detections in 10 min	Recorded negative corpus; not a universal guarantee
Runtime	>=15 Hz fresh valid output; age p95 <=100 ms	10 min hot run; deadline violations and temperature
Loss handling	No publish from samples older than 200 ms	Injected freeze/occlusion; timestamp and wire capture
Optional flight	>=9/10 touchdowns within 0.20 m radial error	All attempts counted; measured vehicle center to pad center

Decision: No current pass is claimed. Repeated-message rate is not fresh-measurement rate.

Test Plan . Bench to Supervised Flight

Progression gates prevent unverified perception from becoming a flight-control input

Stage	Procedure	Exit evidence
A - offline	Replay known/wrong/no-tag scenes; test resize, corner order, NaN and pose flips.	Automated checks, reject counts and annotated examples
B - bench	Props removed; measured poses, all axis signs, 90-degree yaw, power/link restart.	Calibration, grid errors, packets and reconnect behavior
C - SITL	Feed ground-truth-derived targets through the real MAVLink backend; inject latency, loss and dropouts.	Trajectory, mode transitions and target-loss outcomes; synthetic labels
D - static integration	On actual FC, props removed; validate parameters, target acquisition, latency and firmware compatibility.	Parameter exports and received/estimated offsets
E - pilot trials	Healthy baseline hover, Precision Loiter, then 1-3 m landing trials under agreed envelope.	Video, .BIN, CSV and measured touchdown positions

Flight is optional and separately scheduled. Pilot controls mode entry and recovery. Stop on EKF warnings, oscillation, repeated target loss, thermal throttling or uncertain takeover. A simulator pass is not proof of real camera or aircraft performance.

Decision: Advance only when the preceding stage passes; retain failures and aborted attempts in the report.

Illustrative Timing and Error Budget

Figure 4 - Assumed design budget, not measured Nano performance

Pipeline stage	Assumed budget
Frame capture / age	20 ms
Detect + pose	25 ms
Gate + transform	3 ms
Encode + transport	7 ms
Total	55 ms

ILLUSTRATIVE PIPELINE LATENCY



Capture | Detect + pose | Gate | Transport

Measure capture-to-send age and FC response separately. Clock correlation is required for cross-device latency; timestamp availability does not remove buffering.

Why timing affects touchdown

Illustrative lateral speed v = 0.30 m/s and latency tau = 0.10 s yield v*tau = 0.03 m displacement before controller dynamics.

For an assumed fx = 600 px and z = 2 m, one pixel corresponds locally to about 3.3 mm lateral displacement near image center.

For a 0.30 m tag at z = 3 m, projected side is approximately fx*s/z = 60 px when level. This is a geometry estimate, not detection proof.

Calibration, blur, tilt, tag flatness, scale and mount error add bias. Measure the final landing error independently.

Decision: Replace all assumed budgets with distributions from the hot-run bench test.

Limits, Failure Modes and Evidence Package

Expected failure paths must be demonstrated, not hidden behind average accuracy

Limits and mitigations

Glare, shadows, blur, rolling shutter and partial occlusion can break tracking. Test the envelope and suppress uncertain estimates.

A single tag may leave the field of view near touchdown. Record minimum usable height; verify the configured loss behavior below it.

Nano thermal or power issues can stop fresh measurements. Monitor temperature, frame age and link health; exercise restart and brownout on bench.

Vision does not supply obstacle avoidance or a full navigation solution. Moving pads and GPS-denied operation need a separate design.

Planned evidence structure

Per-frame CSV: capture/send times, ID, corners, margin, pose C/B, residual, range, validity, reject reason, CPU temperature and frame drops.

Per-session manifest: firmware hash, parameters, dependency pins, camera mode, K/distortion, extrinsics, tag size and configuration checksums.

Preserve MAVLink capture, autopilot .BIN, raw/annotated clips and measured grid/touchdown coordinates with trial IDs.

Report median, p95, maximum, valid fraction and every abort. Raw datasets remain separate from illustrative figures.

Decision: Accept only a bounded PoC supported by traceable records; do not infer production reliability.

Deliverables, Milestones and Client Review

A concrete engineering engagement with truthful completion language

Proposed delivery milestones

M1 - design: this PDF; confirm hardware, envelope and acceptance gates.

M2 - implementation: source repository, pinned setup guide, calibration tools and replay checks.

M3 - integration: parameter bundle, bench/SITL evidence and issue register.

M4 - optional flight: pilot trials and measured results. Cost/schedule agreed after M1.

Review suggestion - after actual delivery

The client should edit freely and use only claims they personally verified. No review is requested before delivery.

"The delivered technical document gave me a clear architecture for Jetson Nano and ArduPilot precision landing, with specific coordinate transforms, MAVLink requirements and a practical test plan. It clearly distinguished planned work from verified results."

Use this only if the document was delivered and the client agrees; add implementation or flight claims only with evidence.

Technical references - official sources checked 03 Oct 2026

[1] ArduPilot developer: Precision Landing
[2] ArduPilot Copter: Precision Landing and Loiter
[3] MAVLink: Landing Target Protocol
[4] AprilRobotics: AprilTag 3 and pose conventions
[5] OpenCV: PnP and IPPE_SQUARE
[6] NVIDIA: JetPack 4.6.2 platform baseline

Decision: The document is delivered; software, integration and flight evidence remain proposed milestones.